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Knowledge Check: The Erdős–Rényi Model

Practice Assignment • 30 min

Due Oct 21, 11:59 PM IST

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item

1. Which of the following are **true** about the Erdős–Rényi model? 1 / 1 point

- [x] It allows mathematical prediction of many network properties

  Nice work

  Feedback: Correct. The simplicity of the model enables analytical derivation of features like degree distribution and component sizes.

- [ ] It generates networks with high clustering coefficients
- [x] It is useful for analyzing connectivity transitions in networks

  Nice work

  Feedback: Correct. It models the emergence of a giant component and phase transitions well.

- [ ] It accurately mimics real-world community structures

2. What distinguishes the G(n, p) model from the G(n, m) model? 1 / 1 point

- ( ) G(n, p) generates trees, and G(n, m) generates cliques.
- ( ) G(n, p) produces deterministic networks while G(n, m) produces stochastic ones
- (•) G(n, p) adds edges probabilistically while G(n, m) adds a fixed number of random edges
- ( ) G(n, m) always produces sparse graphs, and G(n, p) does not.

  Nice work

  Feedback: Correct. G(n, p) uses a fixed probability to connect each pair, while G(n, m) uses a fixed number of edges.

3. At what value of p does a "giant component" first emerge in an Erdős–Rényi network? 1 / 1 point

- ( ) p=1/n^2
- ( ) p=0.5
- (•) p=1/n
- ( ) p=1

  Nice work

  Feedback: Correct. The threshold for the emergence of a giant component is approximately 1/n.

4. What effect does the rewiring parameter p have in the Small World model? 1 / 1 point

- ( ) It always reduces both clustering and path length
- (•) It helps balance clustering and average distance in networks
- ( ) It increases clustering without affecting path length
- ( ) It converts the network into a tree

  Nice work

  Feedback: Correct. A proper value of p can maintain high clustering with a low average distance.

5. Which of the following are true about the clustering coefficient in the Small World model? 1 / 1 point

- [x] It helps indicate how likely neighbors of a node are to be connected

  Nice work

  Feedback: Correct. The clustering coefficient reflects the density of triangles in the network.

- [x] It is high when the rewiring parameter p is small

  Nice work

  Feedback: Correct. Low values of p preserve high clustering.

- [ ] It remains unchanged regardless of the rewiring parameter
- [ ] It is always zero in real networks

6. Why are network generative models useful for studying real networks? 1 / 1 point

- [x] They help identify mechanisms that might explain real-world network properties

  Nice work

  Feedback: Correct. By mimicking real patterns, models can offer insights into underlying processes.

- [x] They allow scalability for testing algorithms under varied conditions

  Nice work

  Feedback: Correct. Synthetic networks can be generated in large numbers to test robustness.

- [ ] They enforce deterministic behavior in all network simulations
- [ ] They eliminate the need to understand network dynamics

7. What are **some limitations** of the Erdős–Rényi model in representing real world networks? 1 / 1 point

- [x] It lacks realistic clustering

  Nice work

  Feedback: Correct. The clustering coefficient is too low compared to real-world networks.

- [ ] It always contains multiple isolated components
- [x] It fails to generate power-law-like degree distributions

  Nice work

  Feedback: Correct. The degree distribution in ER networks is Poisson, unlike the power law seen in many real systems.

- [ ] It cannot produce large networks

8. Which of the following is a reason for using synthetic networks in algorithm testing? 1 / 1 point

- ( ) Synthetic networks are always more accurate than real ones
- ( ) Real networks always have complete information
- (•) Real networks may be hard to access or gather in large quantities
- ( ) Synthetic networks are static and easier to visualize

  Nice work

  Feedback: Correct. Network generative models provide scalable alternatives when real data is limited.

9. Which statement best describes the degree distribution in large, sparse Erdős–Rényi (ER) networks? 1 / 1 point

- ( ) It is binomial with variance always equal to the mean
- ( ) It is uniformly distributed
- (•) It approximates a Poisson distribution
- ( ) It follows a power law distribution

  Nice work

  Feedback: Correct. In sparse ER networks, the degree distribution is approximately Poisson.

10. What kind of degree distribution is commonly observed in real networks? 1 / 1 point

- ( ) Uniform
- (•) Power-law-like
- ( ) Binomial
- ( ) Exponential

  Nice work

  Feedback: Correct. Real networks often have a long-tailed, power-law-like distribution.

11. What is the clustering coefficient of an Erdős–Rényi network? 1 / 1 point

- ( ) It is 1 for all values of p
- ( ) It increases with the number of nodes
- ( ) It is undefined due to its stochastic nature
- (•) It is equal to p, the edge probability

  Nice work

  Feedback: Correct. In ER networks, the clustering coefficient equals the probability of edge formation.